Calculus

1. Limits

1.1. Indeterminate forms

- $\frac{0}{0}, \frac{\infty}{\infty}, 0 \times \infty, \infty^0, \infty - \infty, 1^\infty, 0^0$

1.2. Evaluating limits

- $\lim_{x \to a} \frac{x^m - a^m}{x^n - a^n} = \left(\frac{m}{n}\right) \cdot a^{m-n}$
- $\lim_{x \to a} \frac{x^n - a^n}{x - a} = na^{n-1}$
- $\lim_{x \to 0} \frac{\sin x}{x} = 1$
- $\lim_{x \to 0} \frac{\tan x}{x} = 1$
- $\lim_{x \to 0} \frac{e^x - 1}{x} = 1$
- $\lim_{x \to 0} \frac{\ln(1 + x)}{x} = 1$
- $\lim_{x \to 0} (1 + x)^{\frac{1}{x}} = e$
- $\lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x = e$
- L'Hôpital's Rule: If the limit is in $\frac{0}{0}$ or $\frac{\infty}{\infty}$ indeterminate form,

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$$

2. Differentiation

2.1. First Principle

- $\frac{d}{dx}(f(x)) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

2.2. Common Properties

- $\frac{d}{dx}(c) = 0$
- $\frac{d}{dx}(c \cdot f(x)) = c \cdot f'(x)$
- $\frac{d}{dx}(f(x) \pm g(x)) = f'(x) \pm g'(x)$

2.3. Product and Quotient Rule

If $u = f(x)$ and $v = g(x)$,

- $\frac{d}{dx}(u \cdot v) = u \cdot \frac{d}{dx}(v) + v \cdot \frac{d}{dx}(u)$
- $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{\left[v \cdot \frac{d}{dx}(u)\right] - \left[u \cdot \frac{d}{dx}(v)\right]}{v^2}$

2.4. Chain Rule

- $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$

2.5. Power Rule

- $\frac{d}{dx}(x^n) = nx^{n-1}$

2.6. Common Derivatives

1. $\frac{d}{dx}(x) = 1$
2. $\frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}$
3. $\frac{d}{dx}\left(\sqrt{x}\right) = \frac{1}{2\sqrt{x}}$
4. $\frac{d}{dx}(a^x) = a^x \ln a$
5. $\frac{d}{dx}(e^x) = e^x$
6. $\frac{d}{dx}(e^{mx}) = me^{mx}$
7. $\frac{d}{dx}(\ln|x|) = \frac{1}{x}$
8. $\frac{d}{dx}\left(\log_a x\right) = \frac{1}{x \ln a}$

2.7. Derivatives of trig functions

1. $\frac{d}{dx}(\sin x) = \cos x$

2. $\frac{d}{dx}(\cos x) = -\sin x$
3. $\frac{d}{dx}(\tan x) = \sec^2 x$
4. $\frac{d}{dx}(\cot x) = -\csc^2 x$
5. $\frac{d}{dx}(\sec x) = \sec x \tan x$
6. $\frac{d}{dx}(\csc x) = -\csc x \cot x$

2.8. Derivatives of inverse trig functions

1. $\frac{d}{dx}\left(\sin^{-1} x\right) = \frac{1}{\sqrt{1 - x^2}}$
2. $\frac{d}{dx}\left(\cos^{-1} x\right) = \frac{-1}{\sqrt{1 - x^2}}$
3. $\frac{d}{dx}\left(\tan^{-1} x\right) = \frac{1}{1 + x^2}$
4. $\frac{d}{dx}\left(\cot^{-1} x\right) = \frac{-1}{1 + x^2}$
5. $\frac{d}{dx}\left(\sec^{-1} x\right) = \frac{1}{|x|\sqrt{x^2 - 1}}$
6. $\frac{d}{dx}\left(\csc^{-1} x\right) = \frac{-1}{|x|\sqrt{x^2 - 1}}$

2.9. Derivatives of hyperbolic functions

1. $\frac{d}{dx}(\sinh x) = \cosh x$

2. $\frac{d}{dx}(\cosh x) = \sinh x$

3. $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$

4. $\frac{d}{dx}(\coth x) = -\operatorname{csch}^2 x$

5. $\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$

6. $\frac{d}{dx}(\operatorname{csch} x) = -\operatorname{csch} x \coth x$

2.10. Derivatives of inverse hyperbolic functions

1. $\frac{d}{dx}\left(\sinh^{-1} x\right) = \frac{1}{\sqrt{x^2 + 1}}$

2. $\frac{d}{dx}\left(\cosh^{-1} x\right) = \frac{1}{\sqrt{x^2 - 1}}$

3. $\frac{d}{dx}\left(\tanh^{-1} x\right) = \frac{1}{1 - x^2}$

4. $\frac{d}{dx}\left(\coth^{-1} x\right) = \frac{1}{1 - x^2}$

5. $\frac{d}{dx}\left(\operatorname{sech}^{-1} x\right) = \frac{-1}{x\sqrt{1 - x^2}}$

6. $\frac{d}{dx}\left(\operatorname{csch}^{-1} x\right) = \frac{-1}{|x|\sqrt{1 + x^2}}$

3. Integration

3.1. Common Properties

- $\int c dx = cx + C$
- $\int c \cdot f(x) dx = c \int f(x) dx$
- $\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$

3.2. Power Rule

- $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

3.3. Common Integrals

1. $\int \frac{1}{x} dx = \ln|x| + C$
2. $\int \frac{1}{\sqrt{x}} dx = 2\sqrt{x} + C$
3. $\int e^x dx = e^x + C$
4. $\int e^{mx} dx = \frac{1}{m} e^{mx} + C$
5. $\int a^x dx = \frac{a^x}{\ln a} + C$
6. $\int \ln x dx = x \ln x - x + C$

3.4. Integrals of trig functions

1. $\int \sin x dx = -\cos x + C$
2. $\int \cos x dx = \sin x + C$
3. $\int \tan x dx = \ln|\sec x| + C$
4. $\int \cot x dx = \ln|\sin x| + C$
5. $\int \sec x dx = \ln|\sec x + \tan x| + C$
6. $\int \csc x dx = \ln|\csc x - \cot x| + C = \ln\left|\tan \frac{x}{2}\right| + C$
7. $\int \sec^2 x dx = \tan x + C$
8. $\int \csc^2 x dx = -\cot x + C$
9. $\int \sec x \tan x dx = \sec x + C$
10. $\int \csc x \cot x dx = -\csc x + C$

3.5. Integrals of inverse trig functions

1. $\int \sin^{-1} x dx = x \sin^{-1} x + \sqrt{1 - x^2} + C$
2. $\int \cos^{-1} x dx = x \cos^{-1} x - \sqrt{1 - x^2} + C$
3. $\int \tan^{-1} x dx = x \tan^{-1} x - \frac{1}{2} \ln\left|1 + x^2\right| + C$
4. $\int \cot^{-1} x dx = x \cot^{-1} x + \frac{1}{2} \ln\left|1 + x^2\right| + C$
5. $\int \sec^{-1} x dx = x \sec^{-1} x - \ln\left|x + \sqrt{x^2 - 1}\right| + C$

6. $\int \csc^{-1} x dx = x \csc^{-1} x + \ln\left|x + \sqrt{x^2 - 1}\right| + C$

3.6. Integrals of hyperbolic functions

1. $\int \sinh x dx = \cosh x + C$
2. $\int \cosh x dx = \sinh x + C$
3. $\int \tanh x dx = \ln|\cosh x| + C$
4. $\int \coth x dx = \ln|\sinh x| + C$
5. $\int \operatorname{sech} x dx = \tan^{-1}(\sinh x) + C$
6. $\int \operatorname{csch} x dx = \ln\left|\tanh \frac{x}{2}\right| + C$

3.7. Integrals of inverse hyperbolic functions

1. $\int \sinh^{-1} x dx = x \sinh^{-1} x - \sqrt{x^2 + 1} + C$
2. $\int \cosh^{-1} x dx = x \cosh^{-1} x - \sqrt{x^2 - 1} + C$
3. $\int \tanh^{-1} x dx = x \tanh^{-1} x + \frac{1}{2} \ln\left|1 - x^2\right| + C$
4. $\int \coth^{-1} x dx = x \coth^{-1} x + \frac{1}{2} \ln\left|x^2 - 1\right| + C$
5. $\int \operatorname{sech}^{-1} x dx = x \operatorname{sech}^{-1} x + 2 \tan^{-1} \sqrt{\frac{1 - x}{1 + x}} + C$
6. $\int \operatorname{csch}^{-1} x dx = x \operatorname{csch}^{-1} x + \sinh^{-1}|x| + C$

3.8. Advanced Integrals

1. $\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$
2. $\int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \ln\left|\frac{a + x}{a - x}\right| + C$
3. $\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \ln\left|\frac{x - a}{x + a}\right| + C$
4. $\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln\left|x + \sqrt{x^2 + a^2}\right| + C = \sinh^{-1} \frac{x}{a} + C$
5. $\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln\left|x + \sqrt{x^2 - a^2}\right| + C = \cosh^{-1} \frac{x}{a} + C, (x > a)$
6. $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C$
7. $\int \sqrt{x^2 + a^2} dx = \frac{x\sqrt{x^2 + a^2}}{2} + \frac{a^2}{2} \ln\left|x + \sqrt{x^2 + a^2}\right| + C$
   $= \frac{x\sqrt{x^2 + a^2}}{2} + \frac{a^2}{2} \sinh^{-1} \frac{x}{a} + C$
8. $\int \sqrt{x^2 - a^2} dx = \frac{x\sqrt{x^2 - a^2}}{2} + \frac{a^2}{2} \ln\left|x + \sqrt{x^2 - a^2}\right| + C$
   $= \frac{x\sqrt{x^2 - a^2}}{2} - \frac{a^2}{2} \cosh^{-1} \frac{x}{a} + C$
9. $\int \sqrt{a^2 - x^2} dx = \frac{x\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C$

3.9. Definite Integrals

- $\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$
- $\int_a^b f(x) dx = -\int_b^a f(x) dx$
- $\int_a^b f(x) dx = \int_a^b f(z) dz$

- $\int_a^b f(x)dx = \int f(a+b-x)dx$
- $\int_a^b f(x)dx = \int_{a-c}^{b-c} f(x+c)dx$
- $\int_a^b f(x)dx = \int_{a+c}^{b+c} f(x-c)dx$
- $\int_a^\infty f(x)dx = \lim_{b\to\infty} \int_a^b f(x)dx$

4. Solving Integrals

4.1. Integration by Parts

- $\int udv = uv - \int vdu$
- $\int u \cdot vdx = u \int vdx - \int \left[\frac{d}{dx}(u) \int vdx\right]dx$

4.2. Shortcuts

- $\int \frac{f'(x)}{f(x)}dx = \ln|f(x)| + C$
- $\int \frac{f'(x)}{\sqrt{f(x)}}dx = 2\sqrt{f(x)} + C$
- $\int e^{mx}[mf(x) + f'(x)]dx = e^{mx}f(x) + C$

5. Series

5.1. Taylor Series

- $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x - a)^n$

5.2. Maclaurin Series

- $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$

5.3. Common Series

1. $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$
2. $e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$
3. $a^x = 1 + \frac{x \ln a}{1!} + \frac{(x \ln a)^2}{2!} + \frac{(x \ln a)^3}{3!} + \dots$
4. $\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, (-1 < x \leq 1)$
5. $\ln(1 - x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} + \dots, (-1 \leq x < 1)$
6. $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$
7. $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$